

Generative AI in Financial Reporting

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Showcase demo

Outline

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Motivation

Why GAI in financial reporting?

- Generative AI (GAI) will likely reshape many financial-reporting tasks; accounting is among the **most exposed** occupations (Eloundou et al. 2024).
- Its core strength is *natural-language generation* — so **report writing** is the natural early step to study.
- Firms face a real tradeoff:
 - ▶ **Benefits** — lower drafting effort, better readability, accuracy, consistency.
 - ▶ **Costs / agency** — errors and litigation, data leakage, tone manipulation, unsupervised employee use.
- Surveys: 15% of U.S. firms had formally adopted GAI in reporting by early 2024, 30% piloting (KPMG 2024).

Research question & contributions

Question

To what extent have firms begun using GAI to *write* financial disclosures through 2024, and how does usage relate to disclosures' linguistic properties?

- ① **Early evidence** on GAI use in a critical, user-facing step of the reporting process — plus public GenScore data for future research.
- ② **Methodological** — a novel validation of a commercial GAI detector *within financial reports*.

Framework

Benefits, costs, and agency conflicts

Benefits

- Automates drafting, organizing, revising, reformatting.
- Writing-partner quality gains; experiments: +18% quality, -40% time (Noy and Zhang 2023).

Costs

- Inaccuracy / hallucination $\Rightarrow$ litigation risk.
- Privacy on third-party servers; employee resistance.

Net expectation

Many firms / employees use GAI in report writing, concentrated **after ChatGPT (Nov 30, 2022)**.

Agency wedge

Whether GAI helps or harms investors depends on alignment of firm, employee, and investor interests.

Predictions

Cross-sectional adoption — GAI use is higher where the cost–benefit tilts toward adoption:

- small firms (fewer resources)
- no IR officer
- recent significant events needing disclosure updates
- low size-adjusted audit fees

Linguistic properties — direction depends on prompts, LLM tendencies, and human editing:

- **Readability** — may improve (summarization) or worsen (obfuscation).
- **Tone** — incentive to positively spin; LLM evidence mixed.
- **Specificity** — GAI may reduce it absent careful prompting.

Method

Measuring GAI: GenScore

- GenScore = GPTZero's estimated probability of GAI authorship (perplexity & burstiness; document and sentence level).
- **Post-GAI** = disclosures from **December 2022** onward; pre-GAI GenScore treated as *measurement error*.

Baseline model (measurement-error-adjusted via the pre-period):

$$GenScore_{it} = \beta_1 Post_t + \text{Firm-FQ FE}_i + \varepsilon_{it}$$

- Coefficient of interest: $\beta_1 > 0$ if firms use GAI post-launch.
- Estimates are a **lower bound** — human editing removes GAI traces from final reports.

Validation: does GPTZero work *in financial reports*?

Prior work does not validate GPTZero in this context, so the paper tests it first (Type I / Type II errors).

Result

GPTZero reliably flags GAI even when only **2.5% of sentences** for **2.5% of firms** are GAI-modified — i.e. just **0.0625% of all text**.

Validation on a balanced panel of year-end reports, 2018–2020 (pre-confound); 500 firms (50 per 2019 market-value decile), 1,500 reports per disclosure type, first 5,000 words each.

Data

Five disclosure types, U.S. firms 2010–2024

Chosen to span styles and GAI incentives:

Disclosure	Style	Year-over-year stickiness
Risk factors (10-Q/K)	structured, legalistic	~79%
MD&A (10-Q/K)	structured	~69%
Conference-call remarks	spoken, casual	~20%
Earnings press releases	informal, persuasive	~62%
IPO filings	(additional analysis)	—

- Sample starts in 2010 to model a long pre-GAI trend.
- GenScore averages **0.55%** across the sample (10-Q/K & PR from WRDS SEC Analytics; calls from Capital IQ).

Results

Main result: firms *are* using GAI

Headline

Up to 4.5% of new disclosure text was GAI-written in 2024

2024 new-text GAI share	Estimate
IPO business descriptions	4.5%
Risk factors	4.3%
MD&A	2.5%

- GenScore rises markedly **after ChatGPT**; significant GAI usage in **all five** disclosure types.
- Reports are heavily human-edited $\Rightarrow$ these are **lower bounds**.

Cross-section: who uses GAI more?

Usage is predictably higher among:

- **Small firms** — the largest gaps: **+292%** in conference calls, **+184%** in press releases vs. large firms.
- Firms with **no investor-relations officer**.
- Firms with **recent significant events** requiring disclosure updates.
- Firms with **low size-adjusted audit fees**.

Adoption highest in technology / communications, lowest in utilities / energy; GAI used most when writing about negative performance and special items.

Linguistic properties of GAI content

Within-report contrast of least- vs. most-likely-GAI sentences:

- **More positively toned** — consistent with spin.
- **More readable** — consistent with clarity gains.
- **More specific** (more numbers / proper nouns) — likely reflects *careful human prompting / editing*.
- **Conference calls**: longer, *no* tone change — relations differ for spoken content.

Caution

GAI choice is endogenous — these are associations, not causal effects.

Magnitudes in context

- Far below the **95%** implied by the Goldman Sachs CEO quote.
- Far below **24%** of text GAI-written in broad corporate press releases (Liang et al. 2025).

Interpretation

Earnings press releases carry *less* GAI than other corporate communications $\Rightarrow$ firms are **more cautious with GAI in financial reporting** than elsewhere.

Conclusion

Conclusion

- Firms have measurably begun using GAI to write disclosures — $\leq 4.5\%$ of new text by 2024, significant in all five types.
- Usage varies **predictably** (small firms, no IR, events, low audit fees); GAI content is more positive, readable, and specific.
- A **validated** early benchmark + public GenScore data to seed future research.

Caveats

Lower-bound estimates; GAI choice is endogenous (associations, not causal); relies on GPTZero's accuracy; sample ends in 2024.

Slides generated by academic-latex-slides from the published paper, as a showcase example.

Appendix — Backup Slides

A1 — GPTZero validation design

- Balanced panel of year-end reports, **2018–2020** (ends pre-2021 to avoid early-LLM confounds; keeps 2020 COVID disclosures).
- **500 firms** = 50 per decile of 2019 market value; **1,500 reports** per disclosure type.
- First **5,000 words** per report (binding for MD&A/RF ~87%, rare for CC/PR).
- Tests Type I (false positive) and Type II (false negative) error of GenScore when known fractions of text are GAI-modified.

A2 — Identifying true usage from measurement error

$$GenScore_{it} = \beta_1 Post_t + \text{Firm-FQ FE}_i + \varepsilon_{it}$$

- Non-zero GenScore *before* Dec 2022 is assumed to be measurement error (conservative).
- Pre-GAI data model and isolate that error from true post-period usage; tests use new (non-sticky) content, orthogonalized to firm–fiscal-quarter FE and controls.

A3 — Adoption & usage texture

- GAI usage largely occurs in *small amounts* within many firms' reports.
- Highest adoption: technology and communications firms; lowest: utilities and energy.
- Firms use GAI most when writing about negative performance growth and special items.
- Context: 23% of accountants used ChatGPT at work by Feb 2024 (Humlum and Vestergaard 2024); 15%/30% adopt/pilot (KPMG 2024).

A4 — Caveats & future research

- Lower bound: only GAI surviving human edits is detected.
- Detected usage may still proxy total usage if strongly correlated.
- Even small GAI use can matter (e.g. data-risk / reputation).
- Future work: cost savings, labor mix, restatements, regulatory scrutiny, market pricing; GAI in other reporting steps; the public GenScore data enable replication and extension.